

Database Management System (DBMS) Notes

Introduction to DBMS

A Database Management System (DBMS) is software used to store, manage, and retrieve data efficiently. It acts as an interface between users and databases. Examples include MySQL, PostgreSQL, Oracle, and MongoDB.

Advantages of DBMS

- Reduces data redundancy
- Ensures data consistency
- Provides data security
- Supports backup and recovery
- Enables multi-user access

Types of Databases

- Hierarchical Database
- Network Database
- Relational Database
- Object-Oriented Database
- NoSQL Database

Relational Model

In the relational model, data is stored in tables called relations. Rows are called tuples and columns are called attributes.

Primary Key and Foreign Key

Primary Key: Uniquely identifies each record in a table.

Foreign Key: Creates a relationship between two tables.

Normalization

Normalization is the process of organizing data to reduce redundancy.

1NF: Eliminate repeating groups

2NF: Remove partial dependency

3NF: Remove transitive dependency

SQL Basics

DDL Commands: CREATE, ALTER, DROP
DML Commands: INSERT, UPDATE, DELETE
DQL Command: SELECT
DCL Commands: GRANT, REVOKE

Joins in SQL

- INNER JOIN
- LEFT JOIN
- RIGHT JOIN
- FULL JOIN

Transactions and ACID Properties

Atomicity: All operations complete or none do.
Consistency: Database remains valid.
Isolation: Transactions do not interfere.
Durability: Changes remain after system failure.

Indexing

Indexes improve database query performance by allowing faster data retrieval.

ER Model

Entity Relationship (ER) Model represents entities, attributes, and relationships graphically.

Concurrency Control

Concurrency control ensures simultaneous transactions execute safely without conflicts.

DBMS vs File System

DBMS provides better security, consistency, concurrency, and recovery compared to traditional file systems.

Important SQL Commands

Command	Purpose
SELECT	Retrieve data
INSERT	Add new records

UPDATE	Modify existing records
DELETE	Remove records
CREATE	Create database objects